HubPlatform Build Manual

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Get Sources

Get sources from git hub repository:

```
git clone git@github.com:ezloteam/hubPlatform.git --recurse-submodules
```

Check that you have last sources and you chosen master branch. Execute It in the *hubPlatform* folder:

```
git checkout master &&  git pull && git submodule init && git submodule sync --recursive && git submodule
update --recursive
```

For more information you can read: GitHub Workflow

Requirements

- **OS:** Linux amd64
- **kernel** >= 3.4.103
- **libc** >= 2.19
- **cmake** >= 3.5

jsonref

Install **jsonref**

```
sudo apt-get install python3-pip
python3 -m pip install jsonref
```

Nasm

Install **nasm**

```
sudo apt-get install nasm
```

Toolchain Setup

Create folder for toolchaines:

```
sudo mkdir -p /usr/local/toolchains
```

Change permissions for toolchains folder:

```
sudo chown -R $USER:$USER /usr/local/toolchains
```

Create folder for downloaded toolchaines:

```
mkdir -p /usr/local/toolchains/downloads
```

Build instructions(CMake)

CMake version should be higher than 3.13. You can check your version by that command:

```
cmake --version
```

If version of your cmake is less than 3.13 you should execute that (installation of last cmake version):

```
sudo apt remove --purge cmake
hash -r
sudo snap install cmake --classic
```

Recommended structure for build folders:

```
<SOURCES>/build/arm_debug
<SOURCES>/build/amd64_release
```

HubPlatform code can be compiled only with one of toolchains. Use **`CMAKE_TOOLCHAIN_FILE`** cmake flag for generating particular configuration.

Important: If you want to use different toolchain for existing cmake directory, need to remove all content of this folder before generating.

Architecture	Toolchain File
arm	`misc/cmake/arm.toolchain.cmake`
amd64	`misc/cmake/amd64.toolchain.cmake`
mips	`misc/cmake/mips.toolchain.cmake`
arm-openwrt	`misc/cmake/arm-openwrt.toolchain.cmake`

Go to sources folder and create folder for building.

Example for **`amd64`** debug configuration:

```
mkdir build/amd64_debug
cd build/amd64_debug
cmake -DCMAKE_TOOLCHAIN_FILE=misc/cmake/amd64.toolchain.cmake -DHUB_TYPE=local -DCMAKE_BUILD_TYPE=Debug -
DBUILD_TESTS=OFF -DWITH_TESTS=ON -DHUB_DEPLOY_ZIGBEE=OFF -DHUB_MAC=<your_hub_MAC> -
DHUB_SERIAL=<your_hub_serial> ../..
```

For disabling colored cmake output you can add **`HUB_COLORED_OUTPUT=OFF`** flag:

```
cmake -DCMAKE_TOOLCHAIN_FILE=misc/cmake/amd64.toolchain.cmake -DCMAKE_BUILD_TYPE=Debug -DHUB_COLORED_OUTPUT=OFF
../..
```

Other flags:

Flag	Value Type	Default Value	Description
`WITH_FIRMWARE`	BOOL	`ON`	Need to build firmware projects

WITH_FIRMWARE_TESTS	BOOL	OFF	Need to build tests
IS_NEED_COVERAGE	BOOL	OFF	Need to create target for tests coverage
HUB_COLORED_OUTPUT	BOOL	ON	Use colored output for configuration messages
HUB_TYPE	STRING	a20.1	Type of a hub for deployment (values: a20.1, h2.1 or local)
HUB_ARM_DEPLOY_IP	STRING	127.0.0.1	A20 Hub IP for deploying firmware (for local deployment may be omitted)
HUB_ARM_DEPLOY_PORT	STRING	54345	Port of ezlo hub for deploying firmware (for local deployment may be omitted)

Run unit tests

For running tests could be used `ctest` tool. `ctest` should be executed in the build folder. Examples:

Command	Description
`ctest`	Run all existing tests
`ctest -j2`	Run all existing tests in 2 threads
`ctest --output-on-failure`	Show test output if it's failed
`ctest -R lua`	Run tests modules which has `lua` word in the name
`ctest -R lua -VV`	Run tests modules which has `lua` word in the name and show all tests output

Deploy on A20 hub

ARM binaries could be deployed on A20 hubs by `deploy` target. Add `cmake` flag with your hub ip:

```
cmake -DCMAKE_TOOLCHAIN_FILE=misc/cmake/arm.toolchain.cmake -DHUB_TYPE=a20.1 -DHUB_ARM_DEPLOY_IP=192.168.11.1 -
DHUB_ARM_DEPLOY_ROOT_PASSWORD="<password>" ../..
```

After building you can run `deploy` target:

```
make deploy
```

Deploy on H2 hub

Refer to Deploy and test firmware on H2